

Лабораторная работа №4

Настройка HTTP-сервера Apache и виртуального хостинга

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Цель работы

Получить практические навыки установки и настройки **Apache HTTP Server**, включая работу с DNS-зонами и конфигурацией **виртуального хостинга**.

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Конфигурация HTTP-сервера

Просмотр конфигурационных файлов

```

Installing packages...
Installed:
apr-1.7.5-2.el10.x86_64
apr-util-lmdb-1.6.3-21.el10.x86_64
httpd-2.4.63-1.el10_0.2.x86_64
httpd-fsfilesystem-2.4.63-1.el10_0.2.noarch
httpd-tools-2.4.63-1.el10_0.2.x86_64
mod_http2-2.0.29-2.el10_0.1.x86_64
mod_ssl-1:2.4.63-1.el10_0.2.x86_64
apr-util-1.6.3-21.el10.x86_64
apr-util-openssl-1.6.3-21.el10.x86_64
httpd-core-2.4.63-1.el10_0.2.x86_64
httpd-manual-2.4.63-1.el10_0.2.noarch
mod_fcgid-2.3.9-35.el10.x86_64
mod_lua-2.4.63-1.el10_0.2.x86_64
rocky-logos-httpd-100.4-7.el10.noarch

Complete!
[root@server.vabazlov.net ~]# ls /etc/httpd/conf
httpd.conf  magic
[root@server.vabazlov.net ~]# ls /etc/httpd/conf.d
autoindex.conf fcgid.conf manual.conf README ssl.conf userdir.conf welcome.conf
[root@server.vabazlov.net ~]#
```

Рис. 1: Конфигурационные файлы

```
-trapper zaobix-web-service zero-k zerotier
[root@server.vabazlov.net ~]# firewall-cmd --add-service=http
success
[root@server.vabazlov.net ~]# firewall-cmd --add-service=http --permanent
success
[root@server.vabazlov.net ~]# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
[root@server.vabazlov.net ~]# systemctl start httpd
[root@server.vabazlov.net ~]# █
```

Рис. 2: Запуск httpd

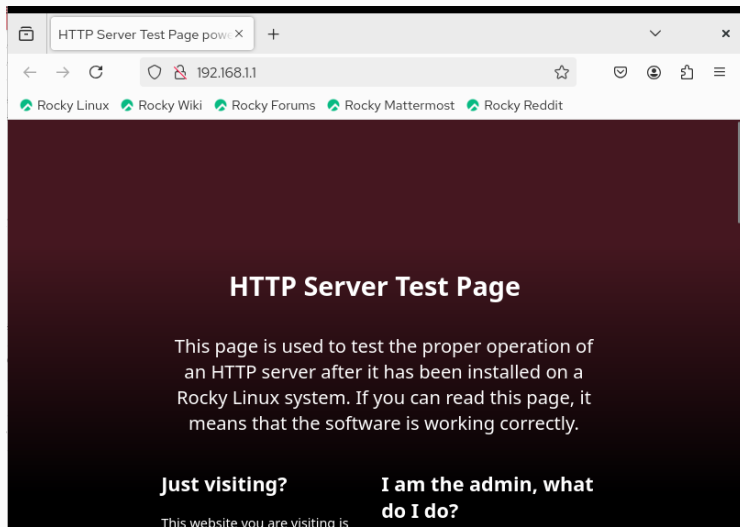
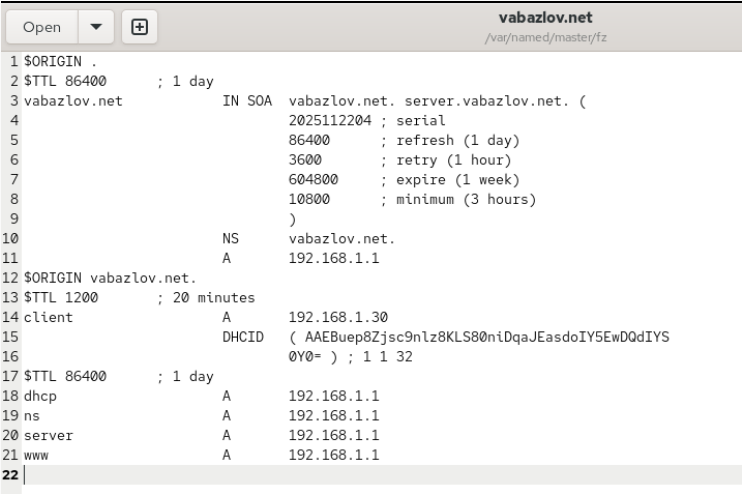


Рис. 3: Тестовая страница Apache

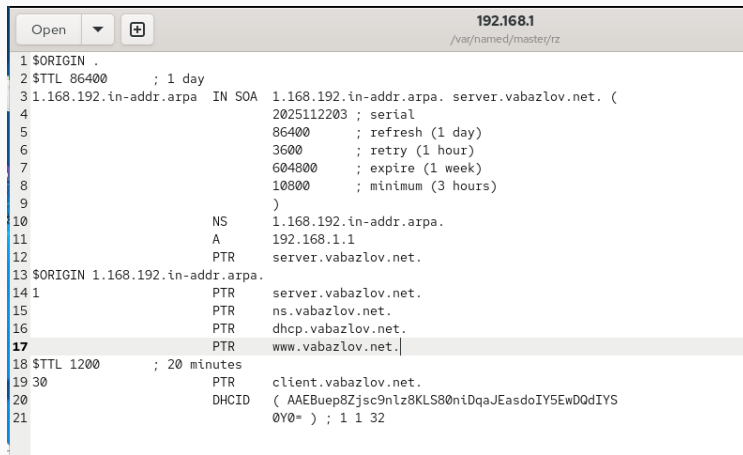
```
[vabazlov@server.vabazlov.net ~]$  
[vabazlov@server.vabazlov.net ~]$ sudo -i  
[sudo] password for vabazlov:  
[root@server.vabazlov.net ~]#  
[root@server.vabazlov.net ~]# tail -f /var/log/httpd/error_log  
[Sat Nov 22 10:27:40.136072 2025] [suexec:notice] [pid 13930:tid 13930] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)  
[Sat Nov 22 10:27:40.145242 2025] [lbmethod_heartbeat:notice] [pid 13930:tid 13930] AH02282: No slotmem from mod_heartbeat monitor  
[Sat Nov 22 10:27:40.146673 2025] [systemd:notice] [pid 13930:tid 13930] SELinux policy enabled; httpd running as content system_u:system_r:httpd_t:s0  
[Sat Nov 22 10:27:40.148796 2025] [mpm_event:notice] [pid 13930:tid 13930] AH00489: Apache/2.4.63 (Rocky Linux) OpenSSL/3.2.2 mod_fcgid/2.3.9 configured -- resuming normal operations  
[Sat Nov 22 10:27:40.148806 2025] [core:notice] [pid 13930:tid 13930] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'  
^C  
[root@server.vabazlov.net ~]# tail -f /var/log/httpd/access_log  
192.168.1.30 - - [22/Nov/2025:10:31:21 +0000] "GET / HTTP/1.1" 403 7620 "-" Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"  
192.168.1.30 - - [22/Nov/2025:10:31:21 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15443 "http://192.168.1.1/" Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"  
192.168.1.30 - - [22/Nov/2025:10:31:21 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "http://192.168.1.1/" Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"  
192.168.1.30 - - [22/Nov/2025:10:31:21 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://192.168.1.1/" Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"  
█
```

Рис. 4: Журналы httpd



```
1 $ORIGIN .
2 $TTL 86400      ; 1 day
3 vabazlov.net    IN SOA  vabazlov.net. server.vabazlov.net. (
4                      2025112204 ; serial
5                      86400      ; refresh (1 day)
6                      3600       ; retry (1 hour)
7                      604800     ; expire (1 week)
8                      10800      ; minimum (3 hours)
9                      )
10                 NS     vabazlov.net.
11                 A      192.168.1.1
12 $ORIGIN vabazlov.net.
13 $TTL 1200      ; 20 minutes
14 client         A      192.168.1.30
15                 DHCPID ( AAEBuep8Zjsc9nlz8KLS80niDqaJEasdoIY5EwDQdIYS
16                 0Y0= ) ; 1 1 32
17 $TTL 86400      ; 1 day
18 dhcp           A      192.168.1.1
19 ns             A      192.168.1.1
20 server         A      192.168.1.1
21 www           A      192.168.1.1
22 |
```

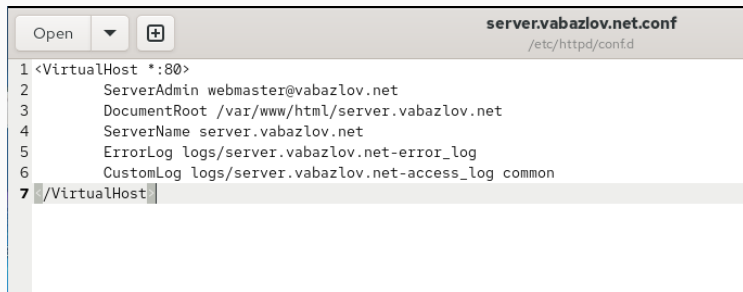
Рис. 5: Прямая зона



The screenshot shows a DNS zone editor interface. At the top, there's a header bar with 'Open', a dropdown arrow, and a '+' icon. To the right of the header, the IP address '192.168.1' is displayed, with the file path '/var/named/master/rz' below it. The main area contains a list of DNS records for a reverse zone. The records are as follows:

```
1 $ORIGIN .
2 $TTL 86400      ; 1 day
3 1.168.192.in-addr.arpa  IN SOA  1.168.192.in-addr.arpa. server.vabazlov.net. (
4                                     2025112203 ; serial
5                                     86400      ; refresh (1 day)
6                                     3600       ; retry (1 hour)
7                                     604800     ; expire (1 week)
8                                     10800      ; minimum (3 hours)
9                                     )
10                                     NS      1.168.192.in-addr.arpa.
11                                     A       192.168.1.1
12                                     PTR     server.vabazlov.net.
13 $ORIGIN 1.168.192.in-addr.arpa.
14 1                                     PTR     server.vabazlov.net.
15                                     PTR     ns.vabazlov.net.
16                                     PTR     dhcp.vabazlov.net.
17                                     PTR     www.vabazlov.net.
18 $TTL 1200      ; 20 minutes
19 30                                     PTR     client.vabazlov.net.
20                                     DHCID   ( AAEBuep8Zjsc9nlz8KLS80niDqaJEasdoIY5EwDQdIYS
21                                     0Y0= ) ; 1 1 32
```

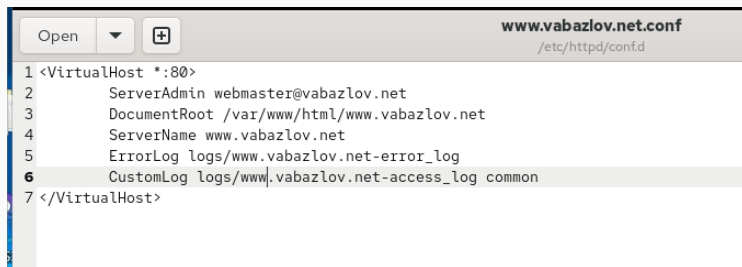
Рис. 6: Обратная зона



The image shows a text editor window with a title bar that includes an 'Open' button, a dropdown arrow, and a '+' icon. The title of the file is 'server.vabazlov.net.conf' and the path is '/etc/httpd/conf.d'. The editor contains the following configuration for a VirtualHost:

```
1 <VirtualHost *:80>
2     ServerAdmin webmaster@vabazlov.net
3     DocumentRoot /var/www/html/server.vabazlov.net
4     ServerName server.vabazlov.net
5     ErrorLog logs/server.vabazlov.net-error_log
6     CustomLog logs/server.vabazlov.net-access_log common
7 /VirtualHost
```

Рис. 7: server.vabazlov.net



The image shows a text editor window with a title bar that reads "www.vabazlov.net.conf" and a subtitle "/etc/httpd/conf.d". The editor contains the following XML configuration for a VirtualHost:

```
1 <VirtualHost *:80>
2     ServerAdmin webmaster@vabazlov.net
3     DocumentRoot /var/www/html/www.vabazlov.net
4     ServerName www.vabazlov.net
5     ErrorLog logs/www.vabazlov.net-error_log
6     CustomLog logs/www.vabazlov.net-access_log common
7 </VirtualHost>
```

Рис. 8: www.vabazlov.net

```
[root@server.vabazlov.net conf.d]# cd /var/www/html
[root@server.vabazlov.net conf.d]# mkdir server.vabazlov.net
[root@server.vabazlov.net conf.d]# cd server.vabazlov.net/
[root@server.vabazlov.net server.vabazlov.net]# touch index.html
[root@server.vabazlov.net server.vabazlov.net]# echo "Welcome to the server.vabazlov.net" > index.html
[root@server.vabazlov.net server.vabazlov.net]# cd ..
[root@server.vabazlov.net html]# mkdir www.vabazlov.net/
[root@server.vabazlov.net html]# cd www.vabazlov.net/
[root@server.vabazlov.net www.vabazlov.net]# echo "Welcome to the www.vabazlov.net" > index.html
[root@server.vabazlov.net www.vabazlov.net]# chown -R apache:apache /var/www
[root@server.vabazlov.net www.vabazlov.net]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.vabazlov.net www.vabazlov.net]# restorecon -vR /var/named
[root@server.vabazlov.net www.vabazlov.net]# restorecon -vR /var/www
[root@server.vabazlov.net www.vabazlov.net]# systemctl restart httpd
[root@server.vabazlov.net www.vabazlov.net]#
```

Рис. 9: Создание контента

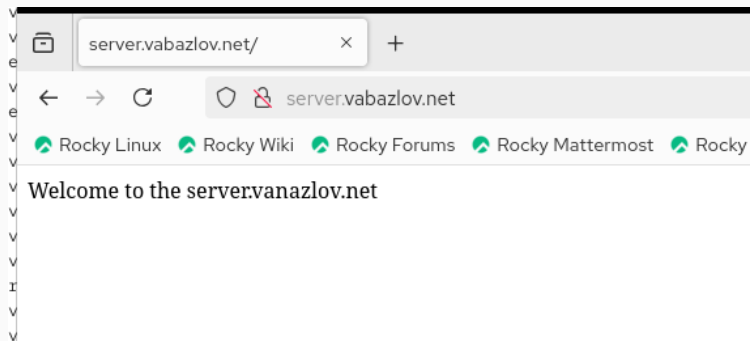


Рис. 10: server.vabazlov.net

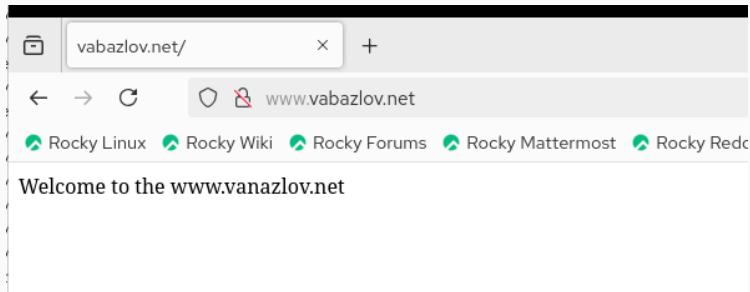


Рис. 11: www.vabazlov.net

Заключение

Настроен веб-сервер Apache с поддержкой виртуального хостинга. Внесены изменения в DNS-зоны, созданы конфигурации хостов и веб-контент, выполнена проверка доступности сайтов. Полученные навыки позволяют развёртывать и сопровождать многосайтовые веб-службы.